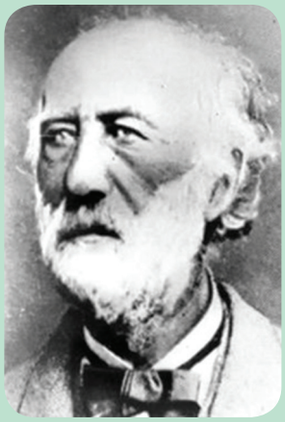
Reinforced concrete:

In 1848, Jean-Louis-Lambot was the first person to use reinforced concrete by using iron bars and wire mesh.





Vibrators are used for this way of compaction. The vibrators permits the use of a lower water- cement ratio as the compaction is very thorough even for drier mixes. The advantages of this type of compaction are,

i. A good surface finish can easily be obtained.
ii. Formworks can be removed early.
iii. A leaner mix with high strength can be obtained.
iv. Concrete can be deposited also in small openings and in the places where it will be difficult to deposit by hand methods.

4.3.7 Curing of Concrete

After concrete is set, it should be continuously cured for a specific period. For the chemical reaction (Hydration) to take place between the constituent of cement, there should be a humidity of 95% in the mix. Hence, it is necessary to keep the concrete wet for a specific period. If the concrete is not cured properly then cracks appear on the top and the full strength also not attained.

4.3.7.1 Methods of Curing

I. Ponding with water.
ii. Covering the concrete with wet sand, jute bags, etc.
iii. Intermittent spraying and covering concrete with polythene bags.
iv. Completely immersing in water tanks as in the case of precast elements.
v. By steam curing.
vi. Vertical surfaces like columns are cured by covering with wet sack or by spraying.



Ponding with water



Intermittent spraying

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